

Proposed Changes to *Empirical Gaussian Processes*

A standalone preview — not part of the paper

Draft for review

August 8, 2026

Summary of proposed changes

Four technical components exist in the implementation but are absent from the manuscript. All four are drafted below as **appendix** material, so the main text grows by only three sentences.

1. **Covariance shrinkage in the M-step** (Appendix A.1). The manuscript’s Conclusion currently lists this as an open limitation (“*possibly requiring shrinkage regularization such as an Inverse-Wishart prior*”), and the derivation is commented out around `main.tex:647`. We show the trace-matched blend is *exactly* MAP-EM under that Inverse-Wishart prior, which turns the limitation into a result.
2. **Base-augmented conditioning** (Appendix A.2). **This one is a reproducibility fix, not an enhancement:** the 7D numbers in `tab:hd_nll` are produced *with* a fitted additive base kernel that the manuscript never mentions. A reader instantiating the plain model will not reproduce that table.
3. **Multi-output Empirical GP** (Appendix A.3) — dense, non-separable cross-output covariance.
4. **Multi-task Empirical GP** (Appendix A.4) — long-format, partial observation, heterogeneous per-task domains.

Section 1 lists the three main-text sentences; Appendix A is the appendix section to paste in wholesale.

1 Main-text edits (three sentences plus one rewrite)

Edit 1 — add after the EM derivation in Section 3

When $K \leq M$ the maximum-likelihood covariance is rank-deficient; we regularize the M-step by shrinking toward a structured target, which is exactly MAP-EM under an Inverse-Wishart prior (Appendix A.1).

Where: Section 3, immediately after the M-step update.

Edit 2 — add where conditioning is introduced

At conditioning time the empirical covariance may be augmented with a fitted additive base kernel, which guarantees the model is never asymptotically worse than that kernel (Appendix A.2).

Where: Section 3, or Section 4.5 with the continuous-domain variant.

Edit 3 — REQUIRED: name the variant behind Table 2

...we evaluate our continuous-domain EM-EGP variant **with a fitted additive base kernel** (Appendix A.2) on a 7-dimensional hyperparameter optimization task from LCBench.

Where: Section 5.5, in the sentence introducing `tab:hd_nll`. Without this the table is not reproducible.

Edit 4 — rewrite one Conclusion sentence

Currently: “With few historical realizations, the empirical covariance is high-variance and can be singular, possibly requiring shrinkage regularization such as an Inverse-Wishart prior.”

Proposed: “With few historical realizations, the empirical covariance is high-variance and can be singular; we address this with a trace-matched shrinkage M-step that is exactly MAP-EM under an Inverse-Wishart prior (Appendix A.1).”

Where: Conclusion, limitations paragraph. This stops us listing as future work something the paper now does.

Edit 5 — OPTIONAL, one sentence for the model family

The same estimator extends to vector-valued and partially-observed multi-task settings (Appendix A.3 and Appendix A.4).

Where: Section 3, or Related Work.

A Model Variants and Regularization

(Paste this whole section into the appendix, after “Additional Empirical Evaluations”.)

A.1 Regularized M-step via Covariance Shrinkage

The M-step update

$$\Sigma_{\text{ML}} = \frac{1}{K-1} \sum_{i=1}^K \left[\mathbf{C}_i + \tilde{\mathbf{m}}_i \tilde{\mathbf{m}}_i^\top \right] \quad (1)$$

is rank-deficient whenever $K \leq M$: the outer products contribute at most $K-1$ directions, and the posterior terms $\mathbf{C}_i$ vanish for completely observed functions. The estimator is therefore singular precisely in the regime the method targets — many reference locations $\mathbf{Z}$, few historical realizations.

Trace-matched shrinkage. We regularize the M-step by blending toward a structured target $\mathbf{B}$ (in practice the base-kernel gram $\mathbf{K}_{\mathbf{Z}\mathbf{Z}}$),

$$\Sigma_{\alpha} = (1 - \alpha) \Sigma_{\text{ML}} + \alpha s \mathbf{B}, \quad s = \frac{\text{tr}(\Sigma_{\text{ML}})}{\text{tr}(\mathbf{B})}, \quad \alpha \in [0, 1]. \quad (2)$$

Trace matching through s makes the blend scale-free: only the *correlation structure* of $\mathbf{B}$ is imposed, never its magnitude, so α carries the same meaning across datasets with different output scales.

Justification as MAP-EM. Equation (2) is not a heuristic. Placing an Inverse-Wishart prior $\Sigma \sim \mathcal{IW}(\Psi, \nu)$ on the covariance, the M-step maximizer of the penalized expected complete-data log-likelihood is

$$\Sigma_{\text{MAP}} = \frac{\Psi + K \Sigma_{\text{ML}}}{K + \nu + M + 1}. \quad (3)$$

Choosing the prior scale $\Psi = (\nu + M + 1) s \mathbf{B}$ recovers (2) exactly, with

$$\alpha = \frac{\nu + M + 1}{K + \nu + M + 1}. \quad (4)$$

So α is an interpretable prior strength, and (4) shows the regularization decays as $O(1/K)$: the estimator is automatically less regularized as historical data accumulates, and $\alpha \rightarrow 0$ recovers maximum likelihood. Because s is re-matched to the data each iteration, the procedure is empirical-Bayes rather than fully Bayesian.

Why a convex blend rather than an additive one. The additive alternative $\Sigma_{\text{ML}} + \alpha \mathbf{B}$ is equally natural but is *not* trace-bounded: every M-step adds $\alpha \text{tr}(\mathbf{B})$, so the iterates can grow without bound and the EM map need not admit a fixed point. The convex blend (2) maps the compact convex set $\{\Sigma \succeq 0 : \text{tr}(\Sigma) \leq T\}$ into itself, so a fixed point exists by Brouwer’s theorem. We therefore use the convex form during *estimation*, and reserve additive augmentation for *conditioning* (Appendix A.2), where the target data — not the EM recursion — controls the added magnitude.

Empirical behavior. Table 1 reports the effect on the LCBench corpus ($M = 200$ reference points, $K = 25$ historical tasks). The unregularized estimator is severely rank-deficient: its effective rank — the exponential of the spectral entropy — is 2.89, with 99.98% of the trace in the leading 22 directions and a median eigenvalue of 2.0×10^{-4} , below the conditioning noise floor. Shrinkage restores conditioning and repairs the resulting overconfidence, raising 95% predictive coverage from 0.65 to 0.98 and reducing predictive NLL by about 8 nats.

Table 1: Effect of M-step shrinkage on the empirical covariance spectrum and on predictive calibration (LCBench, $M = 200$, $K = 25$). Coverage and NLL are measured at acquired points for the frozen-prior variant.

	$\alpha = 0$	$\alpha = 0.3$
Effective rank (of $M = 200$)	2.89	5.05
Trace in top 22 modes	0.9998	0.9624
Median eigenvalue	2.0×10^{-4}	1.6×10^{-2}
95% coverage	0.65	0.98
Predictive NLL	8.30	0.10

Better calibration does not, however, imply better decision-making. When the target task is drawn from the same distribution as the historical corpus, shrinkage *degrades* downstream Bayesian-optimization regret even as it improves calibration, because the empirical covariance is already well matched and the blend only dilutes it. Under distribution shift the sign reverses and shrinkage helps. Shrinkage and target adaptation are therefore substitutes rather than complements, and α is best read as a transfer-distance dial: $\alpha = 0$ in-distribution, $\alpha > 0$ as the target moves away from the corpus.

A.2 Base-Augmented Conditioning

At conditioning time we augment the empirical covariance with a parametric base kernel,

$$k(\mathbf{x}, \mathbf{x}') = k_{\text{emp}}(\mathbf{x}, \mathbf{x}') + k_{\theta}(\mathbf{x}, \mathbf{x}'), \quad (5)$$

fitting θ and the likelihood noise by maximizing the marginal likelihood of the *target* task, with the empirical mean and covariance held fixed. Unlike a convex combination, each component keeps its own magnitude. This gives graceful degradation: when the empirical prior is informative the fitted outputscale of k_{θ} stays small and the model behaves like the Empirical GP; when it is not, k_{θ} grows to explain the target data and the model reverts to a standard GP with that base kernel. The augmented model is therefore never asymptotically worse than its base kernel — a guarantee the raw empirical prior cannot provide at finite rank.

Which parameters are free is a modeling choice with a clear reading. Fitting all of θ suits single-task surrogates; freezing the lengthscales of a pre-fit base kernel and fitting only its outputscale suits transfer, where learned cross-task structure should be reused rather than re-estimated from a handful of target points. We find the plain additive form (5) outperforms richer parameterizations that additionally free the empirical component’s magnitude, which over-parameterize a small target design.

Off-grid augmentation. Care is required in the continuous-domain variant. Recall the interpolated covariance $\Sigma(\mathbf{X}_*, \mathbf{X}_*) = \mathbf{A} + \mathbf{W}\Sigma\mathbf{W}^\top$ with $\mathbf{W} = \mathbf{K}_{*X}\mathbf{K}_{XX}^{-1}$ and Nyström residual $\mathbf{A} = \mathbf{K}_{**} - \mathbf{K}_{*X}\mathbf{K}_{XX}^{-1}\mathbf{K}_{X*}$. The augmentation must add the *full* $k_{\theta}(\mathbf{X}_*, \mathbf{X}_*)$ at the query points, not the projection $\mathbf{W}k_{\theta}(\mathbf{X}, \mathbf{X})\mathbf{W}^\top$ of the base kernel through the reference set; the latter discards exactly the residual variance $\mathbf{A}$ that makes the augmentation useful away from the reference grid. On-grid ($\mathbf{X}_* \subseteq \mathbf{Z}$) the two coincide, which is why the distinction appears only in the continuous-domain setting.

A.3 Multi-Output Empirical Gaussian Processes

The construction extends to vector-valued $f : \mathcal{X} \rightarrow \mathbb{R}^m$ with no extra modeling assumptions, because the empirical estimator is indifferent to how the index set is structured. Given K historical realizations observed on a shared grid of n points for all m outputs, we vectorize each realization, $\text{vec}(\mathbf{Y}_i) \in \mathbb{R}^{nm}$, and form the empirical mean and covariance in the vectorized space. The resulting

$$\Sigma^{\text{MO}} \in \mathbb{R}^{nm \times nm}, \quad [\Sigma^{\text{MO}}]_{(j,a),(l,b)} = \widehat{\text{Cov}}[f^{(a)}(x_j), f^{(b)}(x_l)] \quad (6)$$

is a *dense* cross-output covariance, capturing correlations between different outputs at different inputs directly from data rather than imposing a separable intrinsic-coregionalization structure $\Sigma^{\text{MO}} = \mathbf{B} \otimes \mathbf{K}$. Separability is a strong assumption — it forces every output to share one input correlation pattern — and the empirical construction does not need it. Predictions are returned as a multitask multivariate normal over the nm latent values.

Since Σ^{MO} has rank at most K , when $K < nm$ we factor the vectorized realizations by SVD and propagate the low-rank factors, reducing conditioning cost from $O((nm)^3)$ to $O(K^2nm)$ without approximation.

A.4 Multi-Task Empirical Gaussian Processes

Appendix A.3 assumes every output is observed at every input. Many transfer settings violate this: tasks are observed at different locations, over different domains, and in different numbers. We therefore also provide a *long-format* variant in which an input is a pair (x, t) with $t \in \{1, \dots, T\}$ a task index, so each observation carries its own task label and no completeness assumption is needed.

Historical data may be heterogeneous across tasks: task t contributes curves on its own grid $\mathbf{X}^{(t)} \in \mathbb{R}^{n_t}$, with n_t and the domain differing freely between tasks. The only requirement is that the K realizations are *paired* across tasks — realization i of task t and realization i of task t' arise from the same underlying draw — since this pairing identifies the cross-task covariance

$$\widehat{\text{Cov}}[f^{(t)}(x), f^{(t')}(x')] = \frac{1}{K} \sum_{i=1}^K (f_i^{(t)}(x) - \hat{\mu}^{(t)}(x))(f_i^{(t')}(x') - \hat{\mu}^{(t')}(x')), \quad (7)$$

evaluated by interpolating each task’s empirical basis to the queried locations. Within-task covariance is the one-dimensional empirical kernel of the main text; the cross-task blocks are estimated the same way, so a single estimator supplies the entire $(\sum_t n_t)$ -dimensional joint covariance. This is the variant to use when historical tasks only partially overlap, the common case in learning-curve and hyperparameter-transfer corpora.